INCH-POUND
MIL-M-38510/657
AMENDMENT 7
23 May 2000
SUPERSEDING
MIL-M-38510/657
AMENDMENT 6
17 March 1998

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, HIGH SPEED, CMOS,
BUFFER GATES, MONOLITHIC SILICON

Inactive for new design after 9 August 1996

This amendment forms a part of MIL-M-38510/657, dated 30 September 1986,
and is approved for use by all Departments and Agencies of the Department of Defense

PAGE 1

*1.2.3 Case outlines: Delete in its entirety and substitute with the following:

Outline letter	Package descriptive designator (see MIL-STD-1835)
C	GDIP1-T14 or CDIP2-T14, 14-lead dual-in-line package
E	GDIP1-T16 or CDIP2-T16, 16-lead dual-in-line package
R	GDIP1-T20 or CDIP2-T20, 20-lead dual-in-line package
D	GDIP1-F14 or CDIP2-F14, 14-lead flat package
F	GDIP2-F16 or CDIP3-F16, 16-lead flat package
S	GDIP2-F20 or CDIP3-F20, 20-lead flat package
X	See figure 5
2	CQCC1-N20, 20 terminal leadless chip carrier package

PAGE 3

*3.2.4 Delete and substitute as; "Case outlines. The case outlines shall be as specified in 1.2.3 and as specified on figure 5."

PAGE 9

TABLE I, Additional supply current quiescent per input pin: delete in its entirety and substitute as follows:

"

Additional supply current quiescent per input pin (one unit load)	$I_{CC\Delta}$ 7/	For all inputs, V_{IL} = 0.4 V and V_{IH} = 2.4 V, Test pin at V_{IN} = 2.4 V, When not being tested, control pins at $V_{IH} = V_{CC}$ and V_{IL} = GND, I_O = 0 µA	52,54	5.5 V		1.50	mA
			51,53, 55,60, 61			3.0	

"

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PAGE 11

TABLE I, Output enable time, device types 06, 07, 08, 09, 60 and 61: delete in its entirety and substitute as follows:

"

Output enable time 5/ 6/	t_{PZH}	C_L = 50 pF minimum R_L = 1 kΩ ±10%	06,07	4.5 V	5	51	ns
			08,09	4.5 V	5	44	ns
			60,61	4.5 V	7	53	ns
	t_{PZL}		06,07	4.5 V	5	51	ns
			08,09	4.5 V	5	44	ns
			60,61	4.5 V	10	53	ns

"

PAGE 11

TABLE I, Transition: Delete in its entirety and substitute as follows:

"

Transition 5/ 6/	t_{THL} t_{TLH}	C_L = 50 pF ±10%	01,02	4.5 V	3	20	ns
			03,04, 06-11	4.5 V	2	16	ns
			05	4.5 V	1	16	ns
			51,52	4.5 V	3	20	ns
			53-55, 60,61	4.5 V	2	16	ns

"

PAGE 21

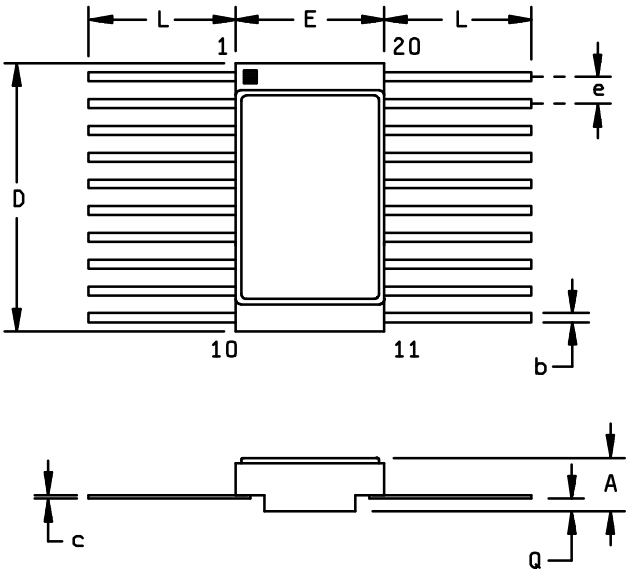
*Figure 1: Delete “Cases R and S” and substitute “Cases R, S, and X”.

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*Add case outline X as FIGURE 5

"Device type 11

Case outline X



Device type 11, case outline X						
Symbol	Inches			Millimeters		
	Min	Nom	Max	Min	Nom	Max
A	.045		.085	1.14		2.16
b	.015		.019	0.38		0.48
c	.003		.006	0.076		0.152
D	.505		.515	12.83		13.08
E	.275		.285	6.99		7.24
e	0.045		0.055	1.14		1.40
L	.250		.370	6.35		9.39
Q	.010			0.25		
N	20			20		

FIGURE 5. Case outline X."

PAGE 49

TABLE III, device type 05, t_{TLH}, T_C = +25°C in the max limits column, delete "2" and substitute "1".
TABLE III, device type 05, t_{TLH}, T_C = +125°C in the max limits column, delete "2" and substitute "1".
TABLE III, device type 05, t_{TLH}, T_C = -55°C in the max limits column, delete "2" and substitute "1".

PAGES 71 THROUGH 75

*Table III: Delete "Cases 2 and R" and substitute "Cases R, S, X, and 2".

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PAGE 76

TABLE III, device type 51, $I_{CC\Delta}$, test number 15: delete in its entirety and substitute as follows:

"

$I_{CC\Delta}$		215	2.4 V		0.4 V		0.4 V		GND		0.4 V		0.4 V		0.4 V	5.5 V	V_{CC}		3.0		3.0		3.0	mA
		216	0.4 V		2.4 V		"		"		"		"		"	"	"		"		"		"	"
		217	"		0.4 V		2.4 V		"		"		"		"	"	"		"		"		"	"
		218	"		"		0.4 V		"		2.4 V		"		"	"	"		"		"		"	"
		219	"		"		"		"		0.4 V		2.4 V		"	"	"		"		"		"	"
		220	"		"		"		"		"		0.4 V		2.4 V	"	"		"		"		"	"

"

PAGE 81

TABLE III, device type 53, $I_{CC\Delta}$, test numbers 23 and 24: delete in its entirety and substitute as follows:

"

ICCΔ		223	2.4 V	0.4 V		0.4 V		0.4 V		0.4 V		GND	0.4 V		0.4 V		0.4 V		0.4 V		GND	5.5 V	VCC		3.0		3.0		3.0	mA
		224	GND	2.4 V		"		"		"		"	"		"		"		"		"	"	"		"		"		"	"
		225	"	0.4 V		2.4 V		"		"		"	"		"		"		"		"	"	"		"		"		"	"
		226	"	"		0.4 V		2.4 V		"		"	"		"		"		"		"	"	"		"		"		"	"
		227	"	"		"		0.4 V		2.4 V		"	"		"		"		"		"	"	"		"		"		"	"
		228	"	'		"		"		0.4 V		"	2.4 V		"		"		"		"	"	"		"		"		"	"
		229	"	"		"		"		"		"	0.4 V		2.4 V		"		"		"	"	"		"		"		"	"
		230	"	"		"		"		"		"	"		0.4 V		2.4 V		"		"	"	"		"		"		'	"
		231	"	"		"		"		"		"	"		"		0.4 V		2.4 V		"	"	"		"		"		"	"
		232	"	"		"		"		"		"	"		"		"		0.4 V		2.4 V	"	"		"		"		"	"

"

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TABLE III, device type 55, $I_{CC\Delta}$, test number 23 and 24: delete in its entirety and substitute as follows:

"

ICCΔ		223	2.4 V	0.4 V		0.4 V		0.4 V		0.4 V		GND	0.4 V		0.4 V		0.4 V		0.4 V		GND	5.5 V	VCC		3.0		3.0		3.0	mA
		224	GND	2.4 V		"		"		"		"	"		"		"		"		"	"	"		"		"		"	"
		225	"	0.4 V		2.4 V		"		"		"	"		"		"		"		"	"	"		"		"		"	"
		226	"	"		0.4 V		2.4 V		"		"	"		"		"		"		"	"	"		"		"		"	"
		227	"	"		"		0.4 V		2.4 V		"	"		"		"		"		"	"	"		"		"		"	"
		228	"	'		"		"		0.4 V		"	2.4 V		"		"		"		"	"	"		"		"		"	"
		229	"	"		"		"		"		"	0.4 V		2.4 V		"		"		"	"	"		"		"		"	"
		230	"	"		"		"		"		"	"		0.4 V		2.4 V		"		"	"	"		"		"		'	"
		231	"	"		"		"		"		"	"		"		0.4 V		2.4 V		"	"	"		"		"		"	"
		232	"	"		"		"		"		"	"		"		"		0.4 V		2.4 V	"	"		"		"		"	"

"

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PAGE 93

TABLE III, device type 60, $I_{CC\Delta}$, test number 23 and 24: delete in its entirety and substitute as follows:

"

ICCΔ		223	2.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	GND								GND	5.5 V	VCC		3.0		3.0		3.0	mA
		224	GND	2.4 V	"	"	"	"	"	"	"	"								"	"	"		"		"		"	"
		225	"	0.4 V	2.4 V	"	"	"	"	"	"	"								"	"	"		"		"		"	"
		226	"	"	0.4 V	2.4 V	"	"	"	"	"	"								"	"	"		"		"		"	"
		227	"	"	"	0.4 V	2.4 V	"	"	"	"	"								"	"	"		"		"		"	"
		228	"	'	"	"	0.4 V	2.4 V	"	"	"	"								"	"	"		"		"		"	"
		229	"	"	"	"	"	0.4 V	2.4 V	"	"	"								"	"	"		"		"		"	"
		230	"	"	"	"	"	"	0.4 V	2.4 V	"	"								"	"	"		"		"		'	"
		231	"	"	"	"	"	"	"	0.4 V	2.4 V	"								"	"	"		"		"		"	"
		232	"	"	"	"	"	"	"	"	0.4 V	"								2.4 V	"	"		"		"		"	"

"

PAGE 95

TABLE III, device type 60, t_{PZH}, test number 133, test limits, T_C = -55°C, min limits column, change from """ to "7".

PAGE 96

TABLE III, device type 60, t_{PZH}, test number 141, test limits, T_C = -55°C, min limits column, change from "10" to "7".

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TABLE III, device type 61, $I_{CC\Delta}$, test number 23 and 24: delete in its entirety and substitute as follows:

"

ICC∆		223	2.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	GND								GND	5.5 V	VCC		3.0		3.0		3.0	mA
		224	GND	2.4 V	"	"	"	"	"	"	"	"								"	"	"		"		"		"	"
		225	"	0.4 V	2.4 V	"	"	"	"	"	"	"								"	"	"		"		"		"	"
		226	"	"	0.4 V	2.4 V	"	"	"	"	"	"								"	"	"		"		"		"	"
		227	"	"	"	0.4 V	2.4 V	"	"	"	"	"								"	"	"		"		"		"	"
		228	"	'	"	"	0.4 V	2.4 V	"	"	"	"								"	"	"		"		"		"	"
		229	"	"	"	"	"	0.4 V	2.4 V	"	"	"								"	"	"		"		"		"	"
		230	"	"	"	"	"	"	0.4 V	2.4 V	"	"								"	"	"		"		"		'	"
		231	"	"	"	"	"	"	"	0.4 V	2.4 V	"								"	"	"		"		"		"	"
		232	"	"	"	"	"	"	"	"	0.4 V	"								2.4 V	"	"		"		"		"	"

"

TABLE III, device type 61, t_{PZH}, test number 133, test limits, T_C = -55°C, min limits column, change from """ to "7".

TABLE III, device type 61, t_{PZH}, test number 141, test limits, T_C = -55°C, min limits column, change from "10" to "7".

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Navy - EC
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DLA-CC

Review activities:
Army - AR, MI, SM
Air Force - 19, 99
Navy - AS, CG, MC, SH

Preparing activity:
DLA - CC

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